

DATABASE SYSTEMS LEARNING PROJECT

EER to Relational Mapping

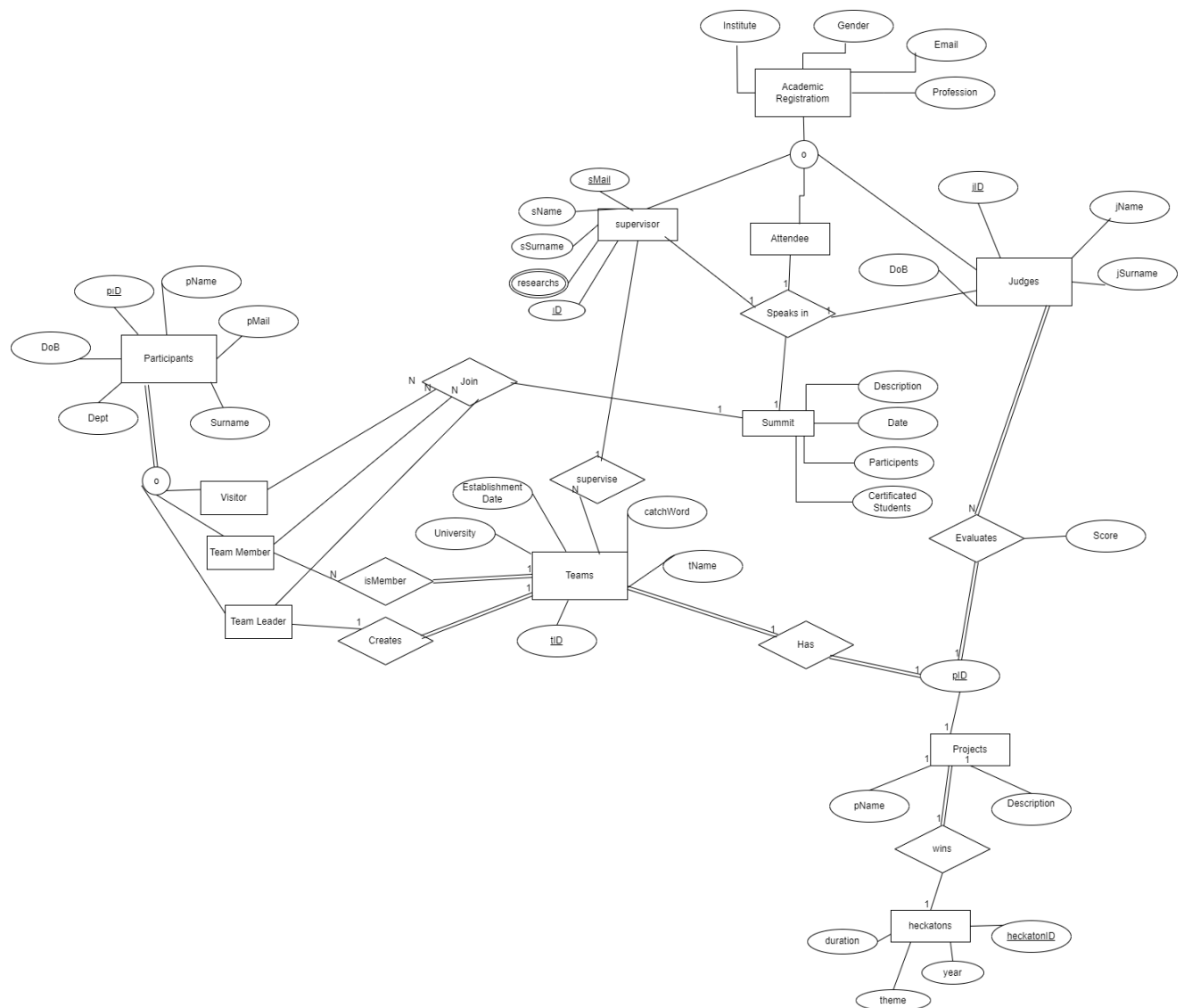
Extended data models and relational schema design

CONTRIBUTORS

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Part1:

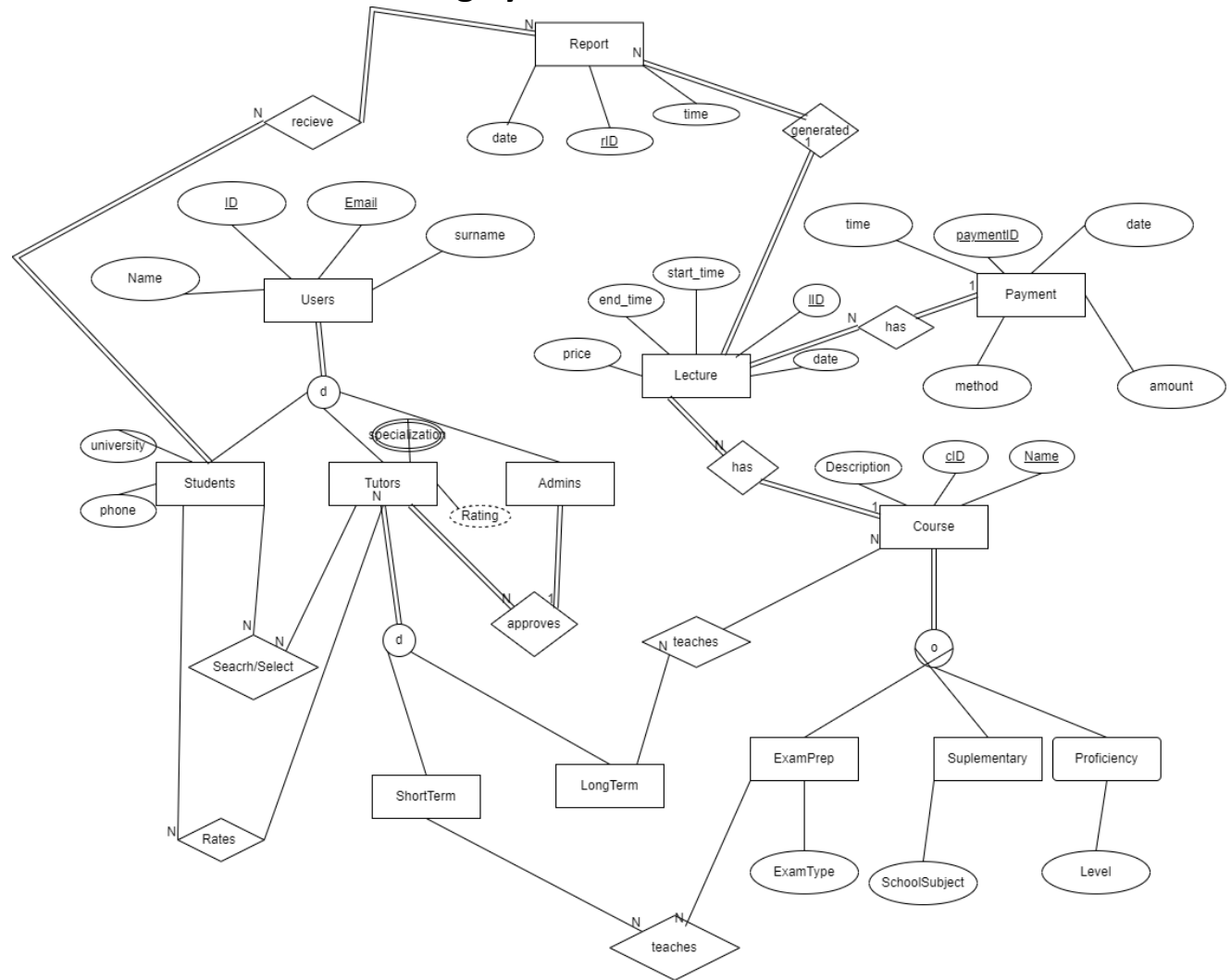
Use Case 1 – ACM Fest Management System



Assumptions:

- teams could not reuse project names so instead of project name being primary key, we add project Id for primary key.

Use Case 2 - Online Tutoring System



- Tutors can have specializations as much as they need.

Part 2:

Use Case 1 – METU NetRegister

Student

<u>StudentID</u>	Name	Surname	Address	PhoneNumber	Password	<u>Email</u>
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Device

<u>MacAddress</u>	Type	Spesifications[FK: Device_Spesifications : Spesifications]	Owner[FK:Student: <u>StudentID</u>]	Model
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MobileDevice

<u>MacAddress</u>	Type	Spesifications[FK: Device_Spesifications : Spesifications]	Owner[FK:Student: <u>StudentID</u>]	OS
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Device_Spesifications

MacAddress[FK: <u>Device</u> : <u>MacAddress</u>]	<u>Specifications</u>
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Registration

Registration_ID	Status	ConnectionType	Day	Time	Device[FK: <u>Device</u> : <u>MacAddress</u>]	ApprovedBy[]
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WirelessRegistration

Registration_ID	Status	ConnectionType	Day	Time	Device[FK: <u>MobileDevice</u> : <u>MacAddress</u>]
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Registration_Location

RegistrationID[FK: <u>Registration</u> : <u>Registration_ID</u>]	<u>Location</u>
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NetworkEngineer

<u>EngineerID</u>	Name	Surname	<u>Email</u>
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Block

Engineer_ID[FK: NetworkEngineer :EngineerID]	RegistrationID[]	Duration	Day	Time
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VPNConnection

<u>StartTimeStamp</u>	RegistrationID[FK: <u>Registration</u> : <u>Registration_ID</u>]	EndTimeStamp
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Issues

<u>Day</u>	<u>Time</u>	RegistrationID[FK: <u>Registration</u> : <u>Registration_ID</u>]	Description	ResolutionStatus	AssignedEngID [FK:NetworkEnginner:EngineerID]
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Use Case 2 – Desire for Velocity

Player

<u>PlayerID</u>	Nickname	Password	EloScore	Credits
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DriftRunway

<u>LobbyID</u>	Quotes	Map
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Cosmetics

<u>CosmeticID</u>	Rarity	Price	PID[FK:Player:PlayeID]	Cname[FK:Car:Name]
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Tournament

<u>TournamentID</u>	Size
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Race

<u>RaceID</u>	Map	Laps
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Car

<u>PID[FK:Player:PlayeID]</u>	Name	Brand	Class	Price	Color	AppliedCosmeticID [FK:Cosmetics:CosmeticID]
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Plays

<u>[FK:DriftRunway:LobbyID]</u>	<u>PID[FK:Player:PlayeID]</u>	HighScore
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Joins

<u>TID[FK:Tournament:TournamentID]</u>	<u>CarName[FK:Car:Name]</u>
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Participates

<u>RID[FK:Race:RaceID]</u>	<u>Cname[FK:Car:Name]</u>	<u>PID[FK:Player:PlayeID]</u>	Duration	Position
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Ranked-Race

<u>RID[FK:Race:RaceID]</u>	EloRange	<u>TID[FK:Tournament:TournamentID]</u>	Laps	Map
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Single-Race

<u>RID[FK:Race:RaceID]</u>	Laps	Map
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Livery

<u>CID[FK:Cosmetics:CosmeticID]</u>	Rarity	Price
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Stickers

<u>CID[FK:Cosmetics:CosmeticID]</u>	Wearfloor	GlareRate	Rarity	Price
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Nemesis

<u>Player[FK:Player:PlayeID]</u>	Nemesis[FK:Player:PlayeID]
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